

Research Report

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Reinforcement Learning Based Dynamic Pricing With Exclusivity-seeking And Strategic Consumers

The Original Model

1. Market Structure

- Two competing firms A and B sell products
- Consumers Purchase one unit of the product in each of two periods
- Firms compete in prices

2. Consumer Preferences

- Consumers are distributed on a hoteling line on $[0,1]$
- Firm A is at 0 ; Firm B at 1
- Consumer located at $\theta \in [0,1]$
- Strategic consumers predict future pricing and change their decisions based on that
- Exclusivity seekers experience more disutility when many other consumers purchase the same product. They don't like when a product is very popular

3. Key Results

- BBP can potentially increase firms profits when consumers are highly strategic or exclusivity-seeking
- When BBP is chosen, consumer surplus may decrease.
- Strategic and exclusivity behaviour act as forces that soften competition in the first period

4. Limitations

- The most important limitation is that only two periods are considered
- In real life scenarios, firms learn how to change prices, but in the paper they follow analytically derived pricing rules
- Dynamic and long run effects are not explored

5. Gaps

There are multiple problems with the original model. real world pricing environments are:

1. Multi period and dynamic
2. Highly heterogeneous across consumers
3. Too complex to be analyzed using closed-form equilibrium

Also in real life, firms do not solve analytical models to make decision about their strategy, instead they learn through experiment and market. Reinforcement learning provides a more natural way to simulate and study these kind of environments.

Our Proposed RL Based Dynamic Pricing Model

Summary of The Model

Overview

- Firms are modeled as reinforcement learning agents
- Consumers are simulated agents with heterogeneous preferences.
- Firms decisions and consumer responses unfold over many periods.

Agents and Environment

Firms

- Two firms repeatedly set prices over time
- Each firm observes past prices, demand and popularity measures.
- Firms maximize long-run profit

Consumers

Consumers' characteristics

- The hotelling location
- Exclusivity factor
- Strategic factor

Consumers choose where to buy in each period based on

- Price
- Product popularity
- Expected future utility

Timing of the game (in each period)

1. Firms post prices
2. Consumers calculate utilities and choose a firm
3. Transactions are made , firms earn their profit
4. Product popularity is updated
5. Firms observe outcomes and update their strategies via RL

The Model In Detail

Environment design

- Framework: implement as a Gym-like environment
- Time: discrete time horizon T (e.g., 20–200 steps to study long-run). Not limited to 2 periods.
- Agents: two firm agents (Firm A, Firm B) controlled by RL policies. Consumers are simulated (rule-based or learned).
- Episode: reset with new consumers' draws; at each timestep firms post prices; consumers buy; update purchase history & exclusivity signals; rewards to firms collected; repeat.

Consumers design

Represents N consumers each with:

- location $\theta \in [0,1]$ (Hotelling preference) — static per consumer
- exclusivity sensitivity y_i drawn from a distribution
- discount factor / strategic level δ_i
- purchase frequency: each period every consumer considers buying 1 unit (like original paper)

Utility Function

Utility for consumer i buying from firm j at time t :

$$U_{ijt} = V - Price_{jt} - \epsilon * \text{distance}(\theta_i, location_j) - y_i d_{jt}$$

Consumers are strategic: they maximize

$$\sum_t^T \delta_i^t U_{ijt}$$

We will have three types of consumers. These consumers approximate the utility functions based on history:

- Myopic: choose immediate max utility.
- Limited lookahead: evaluate utility over next H timesteps given predicted prices (e.g., use last k -step moving average).
- Rational-expectation: use true environment dynamics for perfect foresight

Firms Design: state, actions, rewards

Observations (per firm)

Last K periods of:

- own prices and competitor prices (if observable)
- own demand (number of buyers)
- competitor demand (if public)
- estimated consumer mix (if available): e.g., fraction of high- γ customers
- aggregate popularity statistics (used for exclusivity)

Actions

- Pricing action: choose prices to post for *this period*.
 - Option A (**uniform**): price scalar to all consumers.
 - Option B (**BBP**): set $Price_{old}$ and $Price_{new}$, or even per-customer recommended price.
 - For tractability, action space can be continuous (price in $[0, P_{max}]$) or discrete (grid of prices).

Reward

- Primary:

$$Profit \pi_t = Price * Quantity_{sold}$$

- Optionally include long-term objectives: discounted sum of profits, regularization for fairness, penalty for extreme price discrimination, or include social welfare in multi-objective experiments.